

Лабораторная работа №6

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков установки и конфигурирования СУБД MariaDB

Раздел 2

2. Выполнение лабораторной работы

2.1 Установка MariaDB

- Выполнен переход в режим суперпользователя

```
root@server:~# dnf -y install mariadb mariadb-server
Last metadata expiration check: 2:23:19 ago on Tue 10 Feb 2026 09:59:16 PM UTC.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
mariadb                                x86_64            3:10.5.29-3.el9_7 appstream         1.6 M
mariadb-server                         x86_64            3:10.5.29-3.el9_7 appstream         9.7 M
Installing dependencies:
mariadb-common                        x86_64            3:10.5.29-3.el9_7 appstream          27 k
mariadb-errmsg                        x86_64            3:10.5.29-3.el9_7 appstream          210 k
mysql-selinux                         noarch            1.0.14-1.el9_6   appstream          36 k
perl-DBD-MariaDB                     x86_64            1.21-17.el9      appstream          149 k
perl-Sys-Hostname                    x86_64            1.23-481.1.el9_6 appstream          15 k
Installing weak dependencies:
mariadb-backup                       x86_64            3:10.5.29-3.el9_7 appstream          6.5 M
mariadb-gssapi-server                x86_64            3:10.5.29-3.el9_7 appstream          14 k
mariadb-server-utils                 x86_64            3:10.5.29-3.el9_7 appstream          210 k
Transaction Summary
=====
Install 10 Packages

Total download size: 19 M
Installed size: 113 M
Downloading Packages:
```

2.1 Установка MariaDB

- Выполнен переход в режим суперпользователя
- Установлены пакеты mariadb и mariadb-server через dnf

```
root@server:~# dnf -y install mariadb mariadb-server
Last metadata expiration check: 2:23:19 ago on Tue 10 Feb 2026 09:59:16 PM UTC.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
mariadb                                x86_64            3:10.5.29-3.el9_7 appstream         1.6 M
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Installing dependencies:
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mariadb-errmsg                         x86_64            3:10.5.29-3.el9_7 appstream         210 k
mysql-selinux                         noarch            1.0.14-1.el9_6   appstream         36 k
perl-DBD-MariaDB                      x86_64            1.21-17.el9      appstream         149 k
perl-Sys-Hostname                     x86_64            1.23-481.1.el9_6 appstream         15 k
Installing weak dependencies:
mariadb-backup                        x86_64            3:10.5.29-3.el9_7 appstream         6.5 M
mariadb-gssapi-server                 x86_64            3:10.5.29-3.el9_7 appstream         14 k
mariadb-server-utils                  x86_64            3:10.5.29-3.el9_7 appstream         210 k
Transaction Summary
=====
Install 10 Packages

Total download size: 19 M
Installed size: 113 M
Downloading Packages:
```

2.1 Установка MariaDB

- Выполнен переход в режим суперпользователя
- Установлены пакеты mariadb и mariadb-server через dnf
- Подключены зависимости и компоненты сервера

```
root@server:~# dnf -y install mariadb mariadb-server
Last metadata expiration check: 2:23:19 ago on Tue 10 Feb 2026 09:59:16 PM UTC.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
mariadb                                x86_64            3:10.5.29-3.el9_7 appstream         1.6 M
mariadb-server                         x86_64            3:10.5.29-3.el9_7 appstream         9.7 M
Installing dependencies:
mariadb-common                         x86_64            3:10.5.29-3.el9_7 appstream         27 k
mariadb-errmsg                         x86_64            3:10.5.29-3.el9_7 appstream         210 k
mysql-selinux                          noarch            1.0.14-1.el9_6   appstream         36 k
perl-DBD-MariaDB                      x86_64            1.21-17.el9      appstream         149 k
perl-Sys-Hostname                     x86_64            1.23-481.1.el9_6 appstream         15 k
Installing weak dependencies:
mariadb-backup                         x86_64            3:10.5.29-3.el9_7 appstream         6.5 M
mariadb-gssapi-server                 x86_64            3:10.5.29-3.el9_7 appstream         14 k
mariadb-server-utils                  x86_64            3:10.5.29-3.el9_7 appstream         210 k
Transaction Summary
=====
Install 10 Packages

Total download size: 19 M
Installed size: 113 M
Downloading Packages:
```

- Запущена служба mariadb

```
[root@server.bahis.net ~]# systemctl start mariadb
[root@server.bahis.net ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server.bahis.net ~]# ss -tulpen | grep mysql
[root@server.bahis.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.
```

Рисунок 2: Запуск и включение службы mariadb через systemctl

- Запущена служба mariadb
- Добавлена в автозагрузку (enable)

```
[root@server.bahis.net ~]# systemctl start mariadb
[root@server.bahis.net ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server.bahis.net ~]# ss -tulpen | grep mysql
[root@server.bahis.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.
```

Рисунок 2: Запуск и включение службы mariadb через systemctl

- Запущена служба mariadb
- Добавлена в автозагрузку (enable)
- Созданы символические ссылки systemd

```
[root@server.bahis.net ~]# systemctl start mariadb
[root@server.bahis.net ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server.bahis.net ~]# ss -tulpen | grep mysql
[root@server.bahis.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.
```

Рисунок 2: Запуск и включение службы mariadb через systemctl

- Проверено прослушивание порта 3306

```
[root@server.bahis.net ~]# systemctl start mariadb
[root@server.bahis.net ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server.bahis.net ~]# ss -tulpen | grep mysql
[root@server.bahis.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.
```

Рисунок 3: Проверка прослушивания порта 3306 процессом mariadb

- Проверено прослушивание порта 3306
- Процесс `mariadb` находится в состоянии LISTEN

```
[root@server.bahis.net ~]# systemctl start mariadb
[root@server.bahis.net ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server.bahis.net ~]# ss -tulpen | grep mysql
[root@server.bahis.net ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.
```

Рисунок 3: Проверка прослушивания порта 3306 процессом `mariadb`

- Выполнен `mysql_secure_installation`

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
... Success!
```

- Выполнен `mysql_secure_installation`
- Включена аутентификация `unix_socket`

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
... Success!
```

- Выполнен `mysql_secure_installation`
- Включена аутентификация `unix_socket`
- Установлен пароль `root`

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
... Success!
```

- Выполнен `mysql_secure_installation`
- Включена аутентификация `unix_socket`
- Установлен пароль `root`
- Удалены анонимные пользователи

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
... Success!
```

- Выполнен `mysql_secure_installation`
- Включена аутентификация `unix_socket`
- Установлен пароль `root`
- Удалены анонимные пользователи
- Запрещён удалённый доступ `root`

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
```

- Выполнен `mysql_secure_installation`
- Включена аутентификация `unix_socket`
- Установлен пароль `root`
- Удалены анонимные пользователи
- Запрещён удалённый доступ `root`
- Удалена тестовая БД

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
... Success!
```

remove anonymous users? [Y/n] y

... Success!

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y

... Success!

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

Remove test database and access to it? [Y/n] y

- Dropping test database...

... Success!

- Removing privileges on test database...

... Success!

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

Reload privilege tables now? [Y/n] y

... Success!

Cleaning up...

- Выполнен вход `mysql -u root -p`

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Рисунок 6: Вход в MariaDB под пользователем root

- Выполнен вход `mysql -u root -p`
- Отобразилась версия сервера

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Рисунок 6: Вход в MariaDB под пользователем root

● Просмотр списка клиентских команд

```
MariaDB [(none)]> \h

General information about MariaDB can be found at
http://mariadb.org

List of all client commands:
Note that all text commands must be first on line and end with ';'
?          (\?) Synonym for 'help'.
charset    (\C) Switch to another charset. Might be needed for processing binlog with multi-byte charsets.
clear      (\c) Clear the current input statement.
connect    (\r) Reconnect to the server. Optional arguments are db and host.
delimiter  (\d) Set statement delimiter.
edit       (\e) Edit command with $EDITOR.
ego        (\G) Send command to MariaDB server, display result vertically.
exit       (\q) Exit mysql. Same as quit.
go         (\g) Send command to MariaDB server.
help       (\h) Display this help.
nopager    (\n) Disable pager, print to stdout.
notee      (\t) Don't write into outfile.
nowarning  (\w) Don't show warnings after every statement.
pager      (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print      (\p) Print current command.
prompt     (\R) Change your mysql prompt.
quit       (\q) Quit mysql.
rehash     (\#) Rebuild completion hash.
sandbox    (\-) Disallow commands that access the file system (except \P without an argument and \e).
source     (\.) Execute an SQL script file. Takes a file name as an argument.
status     (\s) Get status information from the server.
system     (\!) Execute a system shell command.
tee        (\T) Set outfile [to_outfile]. Append everything into given outfile.
use        (\u) Use another database. Takes database name as argument.
warnings   (\W) Show warnings after every statement.

For server side help, type 'help contents'
```

Рисунок 7: Список клиентских команд MariaDB

- Выполнен SHOW DATABASES;

```
For server side help, type 'help contents'

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
+-----+
3 rows in set (0.003 sec)

MariaDB [(none)]> exit;
Bye
```

Рисунок 8: Отображение списка системных баз данных (SHOW DATABASES)

- Выполнен SHOW DATABASES ;
- Отображены системные БД

```
For server side help, type 'help contents'

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
+-----+
3 rows in set (0.003 sec)

MariaDB [(none)]> exit;
Bye
```

Рисунок 8: Отображение списка системных баз данных (SHOW DATABASES)

2.2 Конфигурация кодировки символов

- Выполнен вход в MariaDB

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          14
Current database:
Current user:            root@localhost
SSL:                    Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.5.29-MariaDB MariaDB Server
Protocol version:        10
```

2.2 Конфигурация кодировки символов

- Выполнен вход в MariaDB
- Команда `status` показала:

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          14
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.5.29-MariaDB MariaDB Server
Protocol version:       10
```

2.2 Конфигурация кодировки символов

- Выполнен вход в MariaDB
- Команда `status` показала:
- `Server charset — latin1`

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          14
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.5.29-MariaDB MariaDB Server
Protocol version:       10
```


2.2 Конфигурация кодировки символов

- Выполнен вход в MariaDB
- Команда `status` показала:
- `Server charset — latin1`
- `Db charset — latin1`

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          14
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.5.29-MariaDB MariaDB Server
Protocol version:        10
```

2.2 Конфигурация кодировки символов

- Выполнен вход в MariaDB
- Команда `status` показала:
- Server charset — `latin1`
- Db charset — `latin1`
- Client — `utf8`

```
[root@server.bahis.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

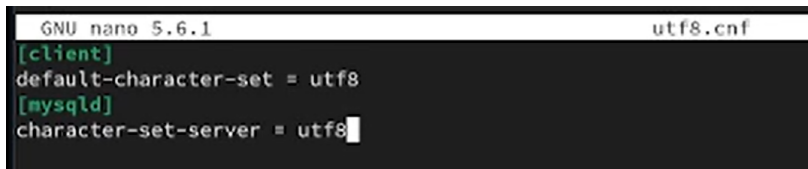
Connection id:          14
Current database:
Current user:            root@localhost
SSL:                    Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.5.29-MariaDB MariaDB Server
Protocol version:        10
```

- Создан файл `/etc/my.cnf.d/utf8.cnf`

```
[root@server.bahis.net ~]# cd /etc/my.cnf.d
[root@server.bahis.net my.cnf.d]# touch utf8.cnf
[root@server.bahis.net my.cnf.d]# nano utf8.cnf
```

Рисунок 10: Создание файла `utf8.cnf` в каталоге `/etc/my.cnf.d`

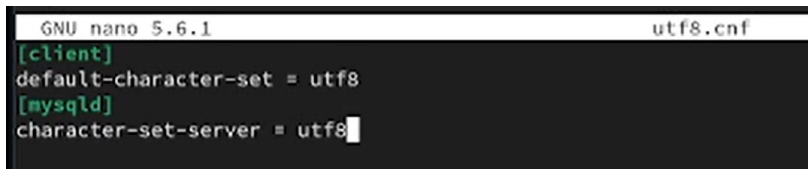
- Добавлены параметры:



```
GNU nano 5.6.1 utf8.cnf
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рисунок 11: Содержимое файла utf8.cnf с параметрами кодировки UTF-8

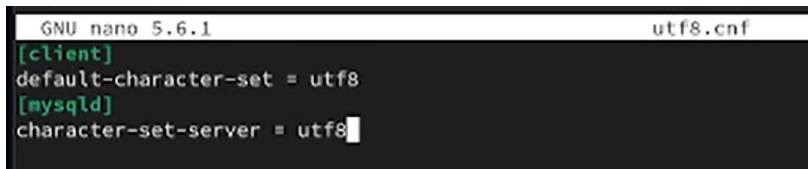
- Добавлены параметры:
- `default-character-set = utf8`



```
GNU nano 5.6.1 utf8.cnf
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рисунок 11: Содержимое файла utf8.cnf с параметрами кодировки UTF-8

- Добавлены параметры:
- `default-character-set = utf8`
- `character-set-server = utf8`



```
GNU nano 5.6.1 utf8.cnf
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рисунок 11: Содержимое файла utf8.cnf с параметрами кодировки UTF-8

- Перезапущена служба mariadb

```
[root@server.bahis.net my.cnf.d]# systemctl restart mariadb
[root@server.bahis.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.5.29-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:      utf8
Db characterset:          utf8
Client characterset:      utf8
Conn. characterset:       utf8
UNIX socket:             /var/lib/mysql/mysql.sock
Uptime:                  40 sec
```

- Перезапущена служба mariadb
- Повторно выполнен status

```
[root@server.bahis.net my.cnf.d]# systemctl restart mariadb
[root@server.bahis.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.5.29-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:      utf8
Db characterset:          utf8
Client characterset:      utf8
Conn. characterset:       utf8
UNIX socket:              /var/lib/mysql/mysql.sock
Uptime:                  40 sec
```


- Перезапущена служба mariadb
- Повторно выполнен status
- Все параметры кодировки — utf8

```
[root@server.bahis.net my.cnf.d]# systemctl restart mariadb
[root@server.bahis.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.5.29-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    utf8
Db characterset:        utf8
Client characterset:    utf8
Conn. characterset:     utf8
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 49 sec
```

2.3 Создание базы данных

- Выполнен вход под root

```
[root@server.bahis.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
```

Рисунок 13: Вход в MariaDB под пользователем root

- Создана БД addressbook (utf8, utf8_general_ci)

```
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.000 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.006 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.004 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.004 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.001 sec)
```

Рисунок 14: Создание базы данных addressbook и проверка списка таблиц

- Создана БД addressbook (utf8, utf8_general_ci)
- Выполнен SHOW TABLES; — таблицы отсутствуют

```
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.000 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.006 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.004 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.004 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.001 sec)
```

Рисунок 14: Создание базы данных addressbook и проверка списка таблиц

- Создана таблица `city` (`VARCHAR(40)`)

```
MariaDB [addressbook]> SELECT * FROM city;
```

name	city
Иванов	Москва
Петров	Сочи
Сидоров	Дубна

```
3 rows in set (0.002 sec)
```

Рисунок 15: Создание таблицы `city`, вставка данных и результат `SELECT * FROM city`

- Создана таблица `city` (`VARCHAR(40)`)
- Добавлены записи: Иванов, Петров, Сидоров

```
MariaDB [addressbook]> SELECT * FROM city;
```

name	city
Иванов	Москва
Петров	Сочи
Сидоров	Дубна

```
3 rows in set (0.002 sec)
```

Рисунок 15: Создание таблицы `city`, вставка данных и результат `SELECT * FROM city`

- Создана таблица `city` (`VARCHAR(40)`)
- Добавлены записи: Иванов, Петров, Сидоров
- Выполнен `SELECT * FROM city;`

```
MariaDB [addressbook]> SELECT * FROM city;
```

name	city
Иванов	Москва
Петров	Сочи
Сидоров	Дубна

```
3 rows in set (0.002 sec)
```

Рисунок 15: Создание таблицы `city`, вставка данных и результат `SELECT * FROM city`

- Создан пользователь bahis@' %'

```
MariaDB [addressbook]> CREATE USER user@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> CREATE USER bahis@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.002 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO bahis@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```


- Создан пользователь `bahis@'%'`
- Назначены права `SELECT`, `INSERT`, `UPDATE`, `DELETE`

```

MariaDB [addressbook]> CREATE USER user@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> CREATE USER bahis@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.002 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO bahis@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

```

- Создан пользователь `bahis@'%'`
- Назначены права `SELECT, INSERT, UPDATE, DELETE`
- Выполнен `FLUSH PRIVILEGES;`

```

MariaDB [addressbook]> CREATE USER user@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> CREATE USER bahis@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.002 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO bahis@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

```

- Создан пользователь `bahis@'%'`
- Назначены права `SELECT, INSERT, UPDATE, DELETE`
- Выполнен `FLUSH PRIVILEGES;`
- Проверена структура таблицы (`DESCRIBE city;`)

```

MariaDB [addressbook]> CREATE USER user@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> CREATE USER bahis@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.002 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.005 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO bahis@'%;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

```

- Проверка через mysqlshow

```
[root@server.bahis.net my.cnf.d]# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
+-----+

[root@server.bahis.net my.cnf.d]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+

[root@server.bahis.net my.cnf.d]# mysqlshow -u bahis -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+

[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
```

- Проверка через `mysqlshow`
- База `addressbook` и таблица `city` доступны

```
[root@server.bahis.net my.cnf.d]# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
+-----+
[root@server.bahis.net my.cnf.d]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
[root@server.bahis.net my.cnf.d]# mysqlshow -u bahis -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
```

2.4 Резервные копии

- Создан каталог /var/backup

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:
- `mysqldump > addressbook.sql`

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```


2.4 Резервные копии

- Создан каталог `/var/backup`
- Выполнен дамп:
- `mysqldump > addressbook.sql`
- Создан сжатый дамп:

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:
- `mysqldump > addressbook.sql`
- Создан сжатый дамп:
- `mysqldump | gzip`

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:
- `mysqldump > addressbook.sql`
- Создан сжатый дамп:
- `mysqldump | gzip`
- Создан дамп с временной меткой

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:
- `mysqldump > addressbook.sql`
- Создан сжатый дамп:
- `mysqldump | gzip`
- Создан дамп с временной меткой
- Выполнено восстановление из .sql

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.4 Резервные копии

- Создан каталог /var/backup
- Выполнен дамп:
- `mysqldump > addressbook.sql`
- Создан сжатый дамп:
- `mysqldump | gzip`
- Создан дамп с временной меткой
- Выполнено восстановление из `.sql`
- Выполнено восстановление из `.sql.gz`

```
[root@server.bahis.net my.cnf.d]# mkdir -p /var/backup
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz
Enter password:
[root@server.bahis.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.bahis.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
```

2.5 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создан каталог `mysql/etc/my.cnf.d`

```
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server.bahis.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mysql.sh
[root@server.bahis.net server]# chmod +x mysql.sh
[root@server.bahis.net server]# mysql.sh
```

Рисунок 19: Создание каталогов `mysql` и копирование конфигурации и резервных копий

2.5 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создан каталог `mysql/etc/my.cnf.d`
- Создан каталог `mysql/var/backup`

```
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server.bahis.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mysql.sh
[root@server.bahis.net server]# chmod +x mysql.sh
[root@server.bahis.net server]# ./mysql.sh
```

Рисунок 19: Создание каталогов `mysql` и копирование конфигурации и резервных копий

2.5 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создан каталог `mysql/etc/my.cnf.d`
- Создан каталог `mysql/var/backup`
- Скопирован `utf8.cnf`

```
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server.bahis.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mysql.sh
[root@server.bahis.net server]# chmod +x mysql.sh
[root@server.bahis.net server]# ./mysql.sh
```

Рисунок 19: Создание каталогов `mysql` и копирование конфигурации и резервных копий

2.5 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создан каталог `mysql/etc/my.cnf.d`
- Создан каталог `mysql/var/backup`
- Скопирован `utf8.cnf`
- Скопированы резервные копии

```
[root@server.bahis.net my.cnf.d]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/
cp: missing destination file operand after '/etc/my.cnf.d/utf8.cnf/vagrant/provision/server/mysql/etc/my.cnf.d/'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server.bahis.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch mysql.sh
[root@server.bahis.net server]# chmod +x mysql.sh
[root@server.bahis.net server]# ./mysql.sh
```

Рисунок 19: Создание каталогов `mysql` и копирование конфигурации и резервных копий

- Создан скрипт `mysql.sh`

```
GNU nano 5.6.1                               mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации
- Создание `/var/backup`

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации
- Создание `/var/backup`
- Запуск и включение `mariadb`

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации
- Создание `/var/backup`
- Запуск и включение `mariadb`
- Неинтерактивный `mysql_secure_installation`

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации
- Создание `/var/backup`
- Запуск и включение `mariadb`
- Неинтерактивный `mysql_secure_installation`
- Создание БД `addressbook`

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```


- Создан скрипт `mysql.sh`
- Установка `mariadb` и `mariadb-server`
- Копирование конфигурации
- Создание `/var/backup`
- Запуск и включение `mariadb`
- Неинтерактивный `mysql_secure_installation`
- Создание БД `addressbook`
- Восстановление из резервной копии

```
GNU nano 5.6.1                                mysql.sh
#!/bin/bash
echo "Provisioning script $0"
systemctl restart named
echo "Install needed packages"
dnf -y install mariadb mariadb-server
echo "Copy configuration files"
cp -R /vagrant/provision/server/mysql/etc/* /etc
mkdir -p /var/backup
cp -R /vagrant/provision/server/mysql/var/backup/* /var/backup
echo "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
then
echo "Securing mariadb"
mysql_secure_installation <<EOF
y
123456
```

- В Vagrantfile добавлен provision-блок

```
server.vm.provision "server mysql",  
type: "shell",  
preserve_order: true,  
path: "provision/server/mysql.sh"
```

Рисунок 21: Добавление конфигурации server mysql в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell

```
server.vm.provision "server mysql",  
type: "shell",  
preserve_order: true,  
path: "provision/server/mysql.sh"
```

Рисунок 21: Добавление конфигурации server mysql в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell
- Путь provision/server/mysql.sh

```
server.vm.provision "server mysql",  
type: "shell",  
preserve_order: true,  
path: "provision/server/mysql.sh"
```

Рисунок 21: Добавление конфигурации server mysql в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell
- Путь provision/server/mysql.sh
- preserve_order: true

```
server.vm.provision "server mysql",  
type: "shell",  
preserve_order: true,  
path: "provision/server/mysql.sh"
```

Рисунок 21: Добавление конфигурации server mysql в Vagrantfile

Раздел 3

3. Выводы

3.1 Выводы

- Установлен и настроен сервер MariaDB

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности
- Настроена кодировка `utf8`

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности
- Настроена кодировка `utf8`
- Создана БД `addressbook` и таблица `city`

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности
- Настроена кодировка `utf8`
- Создана БД `addressbook` и таблица `city`
- Настроены права пользователя

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности
- Настроена кодировка `utf8`
- Создана БД `addressbook` и таблица `city`
- Настроены права пользователя
- Освоены процедуры резервного копирования и восстановления

3.1 Выводы

- Установлен и настроен сервер MariaDB
- Подтверждено прослушивание порта 3306
- Выполнена базовая настройка безопасности
- Настроена кодировка `utf8`
- Создана БД `addressbook` и таблица `city`
- Настроены права пользователя
- Освоены процедуры резервного копирования и восстановления
- Реализована автоматизация через `mysql.sh` и `Vagrantfile`